# INTRODUCTION TO ARTIFICIAL INTELLIGENCE

## III B.TECH - II SEMESTER

| Course Code | Category | Hours/Week | | | Credits | Maximum Marks | | |
|---|---|---|---|---|---|---|---|---|
| | | L | T | P | C | CIE | SEE | Total |
| A6IT39 | PCC | 3 | - | - | 3 | 40 | 60 | 100 |

## COURSE OBJECTIVES

**The course should enable the students to:**

1. State Space Searching techniques to solve a real-world problem
2. Heuristic Searching techniques to optimize a problem iteratively
3. Randomized Search and Emergent Systems to determine the best solution for the developed model
4. Optimal Path algorithms to determine shortest paths in different circuits
5. Problem Decomposition to decompose a problem into manageable sub-components is a necessity in complex problem-solving tasks

## COURSE OUTCOMES

**At the end of the course, student will be able to:**

1. Apply appropriate State Space Searching techniques to solve a real-world problem.
2. Apply appropriate Heuristic Searching techniques to solve a real-world problem.
3. Analyze the problem and infer new knowledge using Randomized Search and Emergent Systems.
4. Apply Optimal Paths Algorithms to real world problems.
5. Analyze the Problem Decomposition Methods and their test cases.

| UNIT - I | INTRODUCTION: ARTIFICIAL INTELLIGENCE | CLASSES: 9 |
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**Introduction:** Artificial Intelligence, Historical Backdrop, What is Artificial Intelligence, AI Agents –types of Agents, Applications of AI.
**State Space Search**: Generate and Test, Simple Search, Depth First Search (DFS), Breadth First Search (BFS), Comparison of BFS and DFS, Depth Bounded DFS (DBDFS), Depth First Iterative Deepening (DFID).

| UNIT - II | HEURISTIC SEARCH | CLASSES: 10 |
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**Heuristic Search:** Heuristic Functions, Best First Search, Hill Climbing, Local Maxima, Solution Space Search, Variable Neighbourhood Descent, Beam Search, Tabu Search, Peak to Peak Methods.

| UNIT - III | RANDOMIZED SEARCH AND EMERGENT SYSTEMS | CLASSES: 10 |
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**Randomized Search and Emergent Systems**: Iterated Hill Climbing, Simulated Annealing, Genetic Algorithms, the Travelling Salesman Problem, Neural Networks, Emergent Systems.

| UNIT - IV | FINDING OPTIMAL PATHS | CLASSES: 10 |
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**Finding Optimal Paths:** Brute Force, Branch and Bound, Refinement Search, Dijkstra's Algorithm, Algorithm A*, Admissibility of A*, Iterative Deepening A* (IDA*), Recursive Best First Search (RBFS), Pruning the Closed List, Pruning the Open List, Divide and Conquer Beam Stack Search.

| UNIT - V | PROBLEM DECOMPOSITION AND CASE STUDIES | CLASSES: 9 |
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**Problem Decomposition:** SAINT, Dendral, Goal Trees, Rule based Systems, XCON, Rule Based Expert Systems.
**Case Studies:** Autonomous Vehicle, Medical Image Analysis.

| TEXT BOOKS |
|---|
| 1. A First Course in Artificial Intelligence, Deepak Khemani, Tata Mc-Graw Hill Education, 2013. |
| **REFERENCE BOOKS** |
| 1. AI – A Modern Approach, Stuart Russel, Peter Norvig, 4th Edition, Pearson Education, 2010.<br>2. Artificial Intelligence, Kevin Knight, Elaine Rich, Shivshankar B. Nair, 3rd Edition, Tata Mc-Graw Hill Education, 2019. |